REPORT

Title

Prolog Program to Implement Monkey Banana Problem

Aim

To implement the **Monkey Banana Problem using Prolog** and find a sequence of actions that allows the monkey to obtain the bananas.

Problem Statement

A monkey is in a room with a box and bananas hanging from the ceiling. The monkey cannot reach the bananas directly. The monkey must move the box below the bananas, climb onto the box, and grasp the bananas.

Requirements

- SWI-Prolog
- Computer system
- Prolog source file (`exp9.pl`)

Procedure

1. Open **SWI-Prolog**.
2. Create a file named `exp9.pl`.
3. Enter the Monkey Banana Problem program.
4. Save the file.
5. Load the program using `[exp9].`.
6. Execute the query `solve(Actions).`.
7. Observe the sequence of actions performed by the monkey.

Program

```
% Monkey Banana Problem

solve(Actions) :-
    solve(monkey(door), box(window), floor, no, Actions).
```

```
solve(monkey(B), box(B), floor, no,
      [push_box(B, middle)|Actions]) :-
    B \= middle,
    solve(monkey(middle), box(middle), floor, no, Actions).

solve(monkey(middle), box(middle), floor, no,
      [climb_box|Actions]) :-
    solve(monkey(middle), box(middle), on_box, no, Actions).

solve(monkey(middle), box(middle), on_box, no,
      [grasp_banana]) :-
    write('Monkey gets the bananas.'), nl.

solve(monkey(A), box(B), floor, no,
      [walk(A,B)|Actions]) :-
    A \= B,
    solve(monkey(B), box(B), floor, no, Actions).
```

Sample Output

Query:

```
?- [exp9].
true.

?- solve(Actions).
```

Output:

```
Monkey gets the bananas.

Actions = [walk(door, window),
            push_box(window, middle),
            climb_box,
            grasp_banana].
```

Conclusion

Thus, the **Monkey Banana Problem was successfully implemented in Prolog**. The program determines a sequence of actions in which the monkey moves to the box, pushes it to the required position, climbs onto it, and finally grasps the bananas.